ClipSAT : Math for AI & Data Science [ANSWER KEY]

Unit 2.01: Neural Layer Matrices & Activations

Q1. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{50} to mathbf{y} in mathbb{R}^{25} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[x] A: 1275
[ ] B: 1250
[ ] C: 75
[ ] D: 1325

Solution: The weight matrix mathbf{W} has shape 25 times 50 , giving 1250 weights.
The bias vector mathbf{b} has 25 parameters. Total = 1250 + 25 = 1275 .

Q2. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{60} to mathbf{y} in mathbb{R}^{30} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[x] A: 1830
[ ] B: 1800
[ ] C: 90
[ ] D: 1890

Solution: The weight matrix mathbf{W} has shape 30 times 60 , giving 1800 weights.
The bias vector mathbf{b} has 30 parameters. Total = 1800 + 30 = 1830 .

Q3. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -4 and z_2 = 6 .

[ ] A: fz_1 = 4, ; fz_2 = 6
[ ] B: fz_1 = -4, ; fz_2 = 6
[ ] C: fz_1 = 0, ; fz_2 = 0
[x] D: fz_1 = 0, ; fz_2 = 6

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and
0 if z le 0 . Thus f-4 = 0 and f6 = 6 .

Q4. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -6 and z_2 = 9 .

[ ] A: fz_1 = 6, ; fz_2 = 9
[ ] B: fz_1 = -6, ; fz_2 = 9
[ ] C: fz_1 = 0, ; fz_2 = 0
[x] D: fz_1 = 0, ; fz_2 = 9

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and
0 if z le 0 . Thus f-6 = 0 and f9 = 9 .

Q5. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -10 and z_2 = 15 .

[ ] A: fz_1 = 0, ; fz_2 = 0
[ ] B: fz_1 = -10, ; fz_2 = 15
[x] C: fz_1 = 0, ; fz_2 = 15
[ ] D: fz_1 = 10, ; fz_2 = 15

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and
0 if z le 0 . Thus f-10 = 0 and f15 = 15 .

Unit 2.01: Neural Layer Matrices & Activations

Q6. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -12 and z_2 = 18 .

[ ] A: fz_1 = 0, ; fz_2 = 0
[ ] B: fz_1 = -12, ; fz_2 = 18
[x] C: fz_1 = 0, ; fz_2 = 18
[ ] D: fz_1 = 12, ; fz_2 = 18

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and 0 if z le 0 . Thus f-12 = 0 and f18 = 18 .

Q7. The sigmoid activation function is sigmaz = frac{1}{1 + e^{-z}} . If sigmaz_0 = 0.20 , what is the value of the derivative sigma'z_0 ?

[x] A: 0.1600
[ ] B: 0.4000
[ ] C: 0.8000
[ ] D: 0.0400

Solution: The derivative of sigmoid satisfies sigma'z = sigmaz1 - sigmaz .
Substituting sigmaz_0 = 0.20 : sigma'z_0 = 0.201 - 0.20 = 0.200.80 = 0.1600 .

Q8. The sigmoid activation function is sigmaz = frac{1}{1 + e^{-z}} . If sigmaz_0 = 0.40 , what is the value of the derivative sigma'z_0 ?

[x] A: 0.2400
[ ] B: 0.8000
[ ] C: 0.6000
[ ] D: 0.1600

Solution: The derivative of sigmoid satisfies sigma'z = sigmaz1 - sigmaz .
Substituting sigmaz_0 = 0.40 : sigma'z_0 = 0.401 - 0.40 = 0.400.60 = 0.2400 .